Society for Computer Technology & Research 's (SCTR's)

Pune Institute of Computer Technology (PICT), Pune

An Autonomous Institute affiliated to the Savitribai Phule Pune University (SPPU)
Approved by AICTE & Government of Maharashtra,
Accredited by NAAC (A+) & NBA [All eligible UG Programs]

Syllabus for

S.Y B. Tech Computer Engineering

(2025-26 Course)

With effect from (AY 2025-26)
National Education Policy (NEP) 2020 Compliant
Approved by the Board of Studies (BoS) and Academic Council

PICT-CE25

Abbreviations :

Sr. No.	Broad Category	Sub- Category of course	Category Code
I.	BSC/ESC	Basic Science Course (BSC)	01
		Engineering Science Course (ESC)	02
II.	Program Courses (PC)	Program Core Course (PCC)	03
		Program Elective Course (PEC)	04
III.	Multidisciplinary courses (MC)	Multidisciplinary Minor (MDM)	05
		Open Elective (OE) Other than particular program	06
IV.	Skill Courses (SC)	Vocational and Skill Enhancement Course (VSEC)	07
V.	Humanities Social Science and Management (HSSM)	Ability Enhancement Course (AEC-01, AEC-02)	08
		Entrepreneurship/Economics/ Management Courses (EEM)	09
		Indian Knowledge System (IKS)	10
		Value Education Course (VEC)	11
VI.	Experiential Learning Courses (ELC)	Research Methodology (RM)	12
		Community engagement Project (CEP) / Field Project (FP)/Co-curricular Activities (CCA)	13
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Detailed guidelines for General Instructions:
Link: General Instructions

Detailed guidelines for Evaluation and Assessment:
Link: Guidelines for Evaluation and Assessment

Detailed guidelines for examination:
Link: Guidelines for examination

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Semester - III

Semester III			Teaching Scheme (Hours/Week)				Credits/ Grades				Examination Scheme and Marks						
											Theory			Practical			
Broad Category of Course	Subject code	Name of subjects	L	P	T	Total	L	P	T	Total	CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE (PR)	ESE (OR)	Total
PCC	1303101	Data Structures (DS)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	1303102	Computer Organisation and Architecture (COA)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	1303103	Discrete Mathematics (DM)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	1303204	Data Structures Lab (DSL)	-	4	-	**4**	–	2	-	**2**	-	-	-	50	50	-	**100**
PCC	1303205	Computer Organization and Architecture Lab (COAL)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	50	**75**
MDM	03051X1	MDM - 1	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
MDM	03053X1	MDM - 1 #	-	-	1	**1**	-	-	1	**1**	-	-	-	25	-	-	**25**
EEM	1309101	Engineering Economics and Financial Management (EEFM)	1	-	-	**1**	1	-	-	**1**	-	-	-	25	-	-	**25**
AEC	0308202	Professional Development and Career Readiness (PDCR)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
OE	03063XX	Foreign Language Studies (FLS)	-	-	2*	**2**	-	-	2	**2**	-	-	-	50	-	-	**50**
VEC	0311101	Universal Human Values (UHV)	2	-	-	**2**	2	-	-	**2**	-	-	-	25	-	-	**25**
CEP	03132XX	Community Engagement Project (CEP)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
Total			**15**	**10**	**2**	**27**	**15**	**5**	**2**	**22**	**80**	**80**	**240**	**250**	**50**	**50**	**750**

L: Lecture, **P:** Practical, **T:** Tutorial,
ISE: In-Semester Examination, **CIE:** Continuous Internal Evaluation, **ESE**: End-Semester Examination,
TW:Term work, **PR:** Practical Examination, **OR**: Oral Examination .

Semester- IV

Semester IV			Teaching Scheme (Hours/Week)				Credits/ Grades				Examination Scheme and Marks						
											Theory			Practical			
Broad Category of Course	Subject code	Name of subjects	L	P	T	Total	L	P	T	Total	CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE (PR)	ESE (OR)	Total
PCC	1403106	Software Engineering (SE)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
PCC	1403107	Database Management Systems (DMS)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	1403108	Operating Systems (OS)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
PCC	1403209	Operating Systems Lab (OSL)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	25	**50**
PCC	1403210	Database Management Systems Lab (DMSL)	-	4	-	**4**	-	2	-	**2**	-	-	-	25	25	-	**50**
MDM	04051X2	MDM - 2	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
MDM	04052X2	MDM - 2 #	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
OE	04063XX	Open Elective - II *	-	-	2	**2***	-	-	2	**2**	-	-	50	-	-	-	**50**
VSEC	1407201	Project Based Learning (PBL)	-	4	-	**4**	-	2	-	**2**	-	-	-	-	-	50	**50**
EEM	1409102	Entrepreneurship (EP)	1	-	-	**1**	1	-	-	**1**	-	-	-	25	-	-	**25**
EEM	1409202	Entrepreneurship Lab (EPL)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
AEC	0408203	Collaborative Skills, Digital Ethics, and Cyber Security (CDC)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
VEC	0411102	Indian Constitution and Social Responsibility (ICSR)	1	-	-	**1**	1	-	-	**1**	-	-	-	25	-	-	**25**
CEP	04132XX	Community Engagement Project (CEP)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
Total			**11**	**18**	**2**	**31**	**11**	**9**	**2**	**22**	**80**	**80**	**290**	**200**	**25**	**75**	**750**

#: Tutorial or laboratory as applicable. Choose one course from MDM Baskets. MDM:X is basket number;. **Refer Annexure-1** for MDM details.

***:** Open Elective (OE) offered by online platforms such as SWAYAM/ NPTEL, **Refer Annexure-II** for details.

XX: Serial number of the courses under that particular category.

Second Year B-Tech
(S. Y B. Tech)
Semester-3

1303101: Data Structures		
Teaching Scheme: L: 03 Hrs./week	**Credits**: 03	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: 1. C Programming for Problem Solving (CPPS)
2. Object Oriented Programming Using C++/Java (OOPC)

Course Objectives:

The objective of this course is to provide the students

1. Basic techniques of algorithm performance analysis.
2. knowledge of various data searching and sorting methods with pros and cons.
3. Structural constraints and advantages in usage of the data using Hashing.
4. Data manipulation using various data structures like Link List, Stack, Queues.
5. Data handling with tree data Structures.

Course Outcomes:

At the end of the course students will be able to

CO1: Analyze the performance of searching, sorting, and hashing techniques.

CO2: Apply linear data structures - Array and Linked lists to store and process structured data.

CO3: Apply principles and primitive operations of Stack and Queue Data Structures to solve the problems like Expression Conversion, Expression Evaluation, and Job scheduling problems.

CO4: Demonstrate basic operations of binary trees and its variants along with their applications.

Course Contents

Module I	Introduction to Analysis of Algorithms	10 Hrs.

Introduction of Algorithms, Analysis of Algorithms, Complexity of algorithms- Space complexity, Time complexity.

Data Structures: Abstract Data Types (ADT), Concept of linear and Non-linear data structures. Use of Array and Associative/Jagged Arrays.

Searching: Search Techniques, Sequential search, Sentinel search, Binary search, Fibonacci search.

Sorting- Types of Sorting- Internal and External Sorting, Stable Sorting, Sorting methods- Bubble sort, Insertion sort, Selection sort, Quick sort, Merge sort, Comparison of All Sorting Methods.

Hashing: Types of Hashing, Hash table, Hash functions, Collision resolution strategies.

Module II	Linked Lists	8 Hrs.

Static memory allocation vs Dynamic Memory Management, linked list using dynamic memory management, Linked List- Abstract Data Type, linked list operations, Types of linked list- Linear, Circular linked lists and Doubly Linked List and operations, doubly circular linked list, Polynomial representation using Linked List, Generalized Linked List (GLL) concept and use of GLL for multi-variable polynomial.

Module III	Stacks & Queues	10 Hrs.
Stacks Abstract Data Type, Representation of Stacks Using Sequential Organization, stack operations, Multiple Stacks, Applications of Stack-Polish notation, need for prefix and Postfix expressions, Expression Evaluation and Conversion, Postfix expression evaluation, Linked Stack and Operations. Recursion- concept, variants of recursion- direct, indirect, head, tail. Queue as Abstract Data Type, Realization of Queues Using Arrays, Circular Queue, Advantages of circular queues. Multi-queues, Deque, Priority Queue, Linked Queue and operations. Applications of Queue- Job Scheduling.		
Module IV	**Trees**	**10 Hrs.**
Definition & Basic Terminology of Tree, Binary Tree, Full Binary Tree, Complete Binary Tree, Binary Search Tree and primitive operations, Threaded Binary Tree, Heap Tree, AVL tree and its operations. Introduction to B Tree & B+ Tree.		

Text Books:

T1. Horowitz and Sahani, "Fundamentals of Data Structures in C++", University Press 2007, 2nd Edition, ISBN- 978-07-1678-292-6.

T2. Goodrich, Tamassia, Goldwasser, "Data Structures and Algorithms in Python", Wiley Publication 2021, ISBN- 978-93-5424-786-6.

T3. Herbert Schildt, "Java: The Complete Reference", McGraw Hill Education, 9th Edition, ISBN- 978-0749467241.

Reference Books:

R1: Brassard & Bratley, "Fundamentals of Algorithmics", Prentice Hall India/Pearson Education 1996, ISBN- 978-8120311312.

R2: R. Gilberg, B. Forouzan, "Data Structures: A Pseudocode approach with C++", Cengage Learn 2005, 2nd Edition, ISBN- 978-8131503140.

R3: M. Weiss, "Data Structures and Algorithm Analysis in C++", 2nd edition, Pearson Education, 2002, ISBN- 978-0201498400.

Relevant Topics for Self-study:

Interpolation Search, Bucket Sort, Shell Sort, Heap Sort, Applications of Linked List, Applications of balanced search trees.

Relevant MOOCs:

1. NPTEL course on, "Programming, Data Structures And Algorithms Using Python", By Prof. Madhavan Mukund **:** **https://onlinecourses.nptel.ac.in/noc22_cs26/preview**
2. NPTEL course on, "Design and analysis of algorithms", By Prof. Madhavan Mukund**:** **https://onlinecourses.nptel.ac.in/noc25_cs23/preview**
3. IITBombayX: Foundations of Data Structures: **https://www.edx.org/learn/data-structures/iitbombay-foundations-of-data-structures**

Other Resources/Links:
1. NPTEL archived course on "**Programming, Data Structures and Algorithms**" by IIT Madras: **https://archive.nptel.ac.in/courses/106/106/106106127/** 2. NPTEL archived course on "**Programming in C++**" by IIT Kharagpur : **https://archive.nptel.ac.in/courses/106/105/106105151/**

1303102: Computer Organization and Architecture		
Teaching Scheme: L: 03 Hrs./week	**Credits**: 03	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Digital Electronics and Logic Design

Course Objectives:
The objective of this course is to provide the students

1. Basics of structure, function , characteristics of computer Organization.
2. Instruction set and logic to build Assembly Language Programs (ALP).
3. Knowledge of instruction level parallelism, parallel organization of multiprocessor,multithreading .
4. Multiple processor organizations , memory architectures (UMA, NUMA, CC-NUMA) and multicore systems.

Course Outcomes:
At the end of the course students will be able to

CO1: Describe the basic components of computer architecture.
CO2: Analyze various addressing modes to perform arithmetic and logical operations using different data types
CO3: Illustrate concept and design principles of memory systems- cache, virtual and main memory.
CO4: Explore multiple processor organizations (SSID, SIMD, MISD, MIMD) and memory architectures.
CO5: Analyze different approaches to multithreading and explore multicore systems.

Course Contents

Module I	Introduction to Processor Architecture	10 Hrs.

Basic structures of computers-Functional units, performance Measurement. Evolution of computer architecture. Processor Structure and Function - Processor Organization, Register Organization, The Instruction cycle, instruction pipeline.

Computer Arithmetic: Integer and Floating-point representation and arithmetic.

Instruction sets: Characteristics and function, Data types, addressing modes and formats, assembler directives, execution of program. Instruction types-data transfer, arithmetic, logical, branch, call, ret.

Reduced instruction set computer: Instruction execution characteristics, RISC architecture, RISC pipeline, RISC Vs CISC.

Module II	Memory organization and management	10 Hrs

Memory system overview: characteristics of memory system, memory hierarchy. Cache Memory – Basics, cache memory principles, elements of cache design, cache coherence and MESI protocol.

Internal Memory Technology: DRAM operation, row decoder column decoder chip organization, error correction code (single and double).

Virtual Memory: main memory allocation, Virtual to Physical Address Translation, Paging, Page Faults, TLB.
Case study: Memory management in advanced processors.

Module III	Parallel processing and Multicore architecture	8 Hrs

Multiple Processor Organization: SISD, SIMD, MISD, MIMD. Uniform memory access (UMA), Non uniform memory access (NUMA), CC-NUMA. Multicore: Hardware and software performance issues, need of multicore. Multicore organization, heterogeneous multicore organization: CPU and GPU.

Case study: Intel Core i7 5960X

Module IV	Multicore programming	8 Hrs.

Software developer viewpoint for multicore programming. Challenges of multicore programming. **Multithreading**: Thread, comparing thread to processes, Approaches of Multithreading.

Communication and synchronization of concurrent tasks: Communication and synchronization, synchronizing concurrency, thread Strategy Approaches

Case study: POSIX 1. Write an application to demonstrate inter process communication.
2. Write a multithreading application for calculators.

Text Books:

T1. W. Stallings, "Computer Organization and Architecture: Designing for Performance",Pearson,11th Edition, 2019, ISBN- 10:0-13-499719-0.

T2. A. Ray, K. Bhurchandi, "Advanced Microprocessors and peripherals: Arch, Programming, Interfacing", Tata McGraw Hill,3rd edition, 2016, ISBN- 978-1259006135.

T3. James Turley, "Advanced 80386 Programming Techniques", McGraw-Hill, 3rd edition, 2015, ISBN-978-0078813429.

T4. Intel® 64 and IA-32 Architectures Software Developer's Manual (Vol. 1-3) - Intel SDM, Intel VT-x & SGX Whitepapers - Virtualization & security documentation.

Reference Books:

R1. C. Hamacher, V. Zvonko, S. Zaky, "Computer Organization", 5th edition, McGraw Hill, 2002, ISBN-978-1259005275.

R2. Cameron Hughes Tracey Hughes "Multicore Programming Design and Implementation for C++ Developers" Wiley,2008, ISBN- 978-0-470-28962-4.

R3. Ray Seyfarth, "Introduction to 64-bit Intel Assembly Language Programming for Linux", 2nd Edition, 2014, ISBN: 1478119209, ISBN- 9781478119203.

R4.https://www.intel.com/content/dam/www/public/us/en/documents/datasheets/core-i7-900-ee-and-desktop-processor-series-datasheet-vol-1.pdf,

Relevant Topics for Self-study:

1. Number System and their conversions.
2. System calls. 3. Handling Procedure and Macros using 64-bit Assembly Language Programming.

Relevant MOOCs:

1. NPTEL course on, "Microprocessor", Dr. Pramod Agarwal, IIT Roorkee, https://nptel.ac.in/courses/108107029/
2. NPTEL course on "C programing and assembly language", Prof. Janakiraman, IIT Madras. https://onlinecourses.nptel.ac.in/noc23_cs93/preview.
3. NPTEL course on, "Computer Architecture", Dr. Indranil Sengupt, Prof. Kamalika Datta, By IIT Kharagpur, https://archive.nptel.ac.in/courses/106/105/106105163/
4. https://archive.nptel.ac.in/courses/106/106/10...

1303103: Discrete Mathematics		
Teaching Scheme: L: 03 Hrs./Week	**Credits:** 03	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Linear Algebra, Basic Mathematics.

Course Objectives:
The objective of this course is to provide the students

1. Fundamentals of set theory and counting principle.
2. Concepts of relations and functions.
3. Role of mathematics in problem solving .
4. Concept of algebraic systems.

Course Outcomes:
At the end of the Course Students will be able to

CO1. Apply formal proof techniques to explain the reasoning clearly.
CO2. Solve the problems of relations and functions.
CO3. Solve the problems-Traveling Salesperson, Minimum Spanning Tree, Shortest Path and Huffman coding using graph theory.
CO4. Evaluate the algebraic structures using properties of binary operations.

Course Contents

Module I	Set Theory and Counting	8 Hrs.

Set Theory: Introduction to Sets– Set Operations, Cardinality of set, Principle of inclusion and exclusion, Types of Sets, Power set, Methods of Proof-Proof by Contradiction, Proof by Mathematical Induction.

Propositional Logic- logic, Propositional Equivalences, Application of Propositional Logic- Translating English Sentences.

Counting: The Basics of Counting, rule of Sum and Product, Permutations and Combinations.

Module II	Relations and Functions	8 Hrs.

Relations: Relations and their Properties, n-ary relations, Representing relations, Equivalence relations, Partial orderings, Hasse diagram, Chains and Antichains, Lattice, Transitive closure and Warshall's algorithm.

Functions: Surjective, Injective and Bijective functions, Identity function, Inverse functions and Compositions of functions.

Module III	Graph Theory and Applications	10 Hrs.

Graph: Graph Terminology and Special Types of Graphs, Representing Graphs and Graph Isomorphism, Euler and Hamilton Paths, the handshaking lemma, Single source shortest path-Dijkstra's Algorithm, Planar Graphs.

Trees: Trees Introduction, properties of trees, Binary tree, Binary search tree, tree traversal, Expression tree, Huffman coding, Minimum Spanning Tree, Kruskal's and Prim's algorithms

Module IV	Algebraic Structures	6 Hrs.
Groups: Algebraic Systems, Semi Groups, Monoids, Groups, Homomorphism and Normal Subgroups. Rings: Integral Domains and Fields, Coding Theory.		

Text Books:

T1. C L Liu, D P Mohapatra, "Elements of Discrete Mathematics", 4th Edition, McGrawHills, ISBN-978-1259006395.

T2. Norman L. Biggs, "Discrete Mathematics", 2nd Edition, Oxford University Press,ISBN- 978-019850717.

Reference Books:

R1. Kenneth H. Rosen,"Discrete Mathematics and its Applications", 8th Edition, McGrawHill, ISBN-978-9390727353.

R2. Bernard Kolman, Robert Busby and Sharon C. Ross, "Discrete Mathematical Structures", 6th Edition, Pearson Education India, ISBN- 978-9332549593.

R3. Sriram Pemmaraju, Steven Skiena, "Computational Discrete Mathematics", 4th Edition, Cambridge University Press, ISBN- 978-1107262904.

Relevant Topics for Self-study :

Implementing Set Operations programmatically, Logical equivalence using a truth table, Application of permutation and combination in binomial theorem. Basic principle of pigeonhole, Graph Traversal Algorithm, Travelling Salesperson problem, Abelian group and its examples.

Relevant MOOCs:

1. https://onlinecourses.nptel.ac.in/noc24_cs92/preview
2. https://onlinecourses.swayam2.ac.in/cec24_ma18/preview

1303204 Data Structures Lab		
Teaching Scheme: P: 04 Hrs./week	**Credits:** 02	**Examination Scheme:** CIE(TW): 50 Marks ESE(PR): 50 Marks

Prerequisite: 1. C Programming for Problem Solving (CPPS)
2. Object Oriented Programming Using C++ (OOPC)

Course Objectives:
The objective of this course is to provide the students

1. Understanding of the working of Searching & Sorting Algorithms.
2. Linked list and types of linked list implementations.
3. Use of Hash Table using Hash Functions.
4. Applications of Stack and Queue data structures.
5. Working of Binary Tree operations.

Course Outcomes:
At the end of the Course Students will be able to
CO1: Implement various Searching & Sorting Algorithms.
CO2: Select appropriate types of linked lists implementations for solving problems.
CO3: Apply suitable Hash Functions for data manipulation in Hash Table.
CO4: Apply Stack and Queue data structures for expression conversion & recursion.
CO5: Implement primitive operations of various types of Binary Trees.

List of Assignments

01.	A customer purchased groceries from a store and wants to verify the prices of specific items in the bill where the items are not in sorted order. Write a C++/Java program to apply any 02 appropriate Searching Algorithms to check the cost of a particular item in the bill. Mention the Time and Space Complexities of Algorithms.
02.	An E-commerce website has a range of products having Product id, name, manufacturer, price and quality rating out of 5. Write a C++/Java Program to display products, 1. In Increasing order of Product id (Use Bubble Sort) 2. In Increasing order of Product price (Use Selection Sort) 3. In Decreasing order of Product Quality Rating (Use Insertion Sort)
03.	Write a C++/Java program to display the employee names in the order of their joining year. Analyze and compare the efficiency and suitability of different algorithms in solving the problem. Justify the choice of algorithms.
04.	Consider a telephone book database where the information is stored as Client's name and telephone number. Write a C++/Java program to quickly search the client's telephone number from the database with O(1) complexity. (Hint: Make use of a Hash table and hash function).
05.	Consider a person creating a playlist of media/music files. The person wants to store music_id, title, time_duration for every entry and wants the following functionalities: 1. Able to insert a media file at any position dynamically. 2. Able to delete any entry from the list dynamically. 3. Able to sort on any field (music_id, title,time_duration)

	4. Able to play continuous looping fashion.
06.	There is a student's club named 'Techno-Fun' club where Students can get membership on request. Similarly one may cancel the membership of a club. First node is reserved for the president of the club and the last node is reserved for the secretary of the club. Choose a type of linked list to store student PRN and Name. Write functions to: a) Add members b) show members c) delete members, d) Compute total number of members e) Two linked lists exist for two divisions. Concatenate two lists.
07.	Browser maintains the history which refers to a list of recently visited websites or web pages in a data structure. Write a C++/Java program to store the browser history or navigation information dynamically as page_id and recent time-stamp. Write functions to… 1. Visit a specific web page. 2. Navigate forward and backward. 3. Add new web pages. 4. Deleting web pages from existing lists.
08.	The ticket booking system of Inox theater has to be implemented using C++/Java program. There are M rows and N seats in each row. Doubly circular linked list has to be maintained to keep track of free seats at rows. Assume some random booking to start with. Use an array to store pointers (Head pointer) to each row. On demand 1. The list of available seats is to be displayed. 2. The seats are to be booked. 3. The booking can be cancelled.
09.	In an application maintaining legal documents which are structured hierarchically with sections, subsections, paragraphs, and sub-paragraphs. Design a program to represent and manage such a document structure. Each node will store either textual data element (title of section, subsection, paragraph, subparagraph), or A pointer to another linked list (subsections, paragraph, subparagraph) Hint: Use Generalized Linked Lists. A = { S1, S2 {P1, P2, P3 {SP1, SP2}}, S3 {P1.P2}, S4, S5}.
10.	In an expression each opening symbol has a corresponding closing symbol and the pairs of parentheses are properly nested. Write a C++/Java program using Stack to check whether a given expression is well parenthesized or not.
11.	The expressions can be represented in infix, prefix and postfix form but the system can evaluate postfix expressions easily. Write C++/Java program for expression conversion as infix to postfix and its evaluation using Stack based on given conditions.
12.	Consider a Pizza parlour where the customers are served from both sides of the waiting line. Choose an appropriate data structure to implement a linear list of elements in which additions and deletions of items may be made at either ends of the list.
13.	Write a C++/Java program to construct a binary search tree by inserting the values in the order given. Perform following operations - i. Insert nodes, ii. Display nodes by Preorder, In-order, Post-order, BFS and DFS. iii. Find number of nodes in longest path from root, iv. Minimum data value found in the tree, v. Change a tree so that the roles of the left and right pointers are swapped at every node, vi. Search for a value.
14.	In the memory the files and folders are stored in Hierarchical Manner. Write a C++/Java program to create the hierarchical structure of files and folders using Binary Tree.

1303205 Computer Organization and Architecture Lab		
Teaching Scheme: P: 02 Hrs./ Week	**Credit**: 01	**Examination Scheme:** CIE (TW): 25 Marks ESE(OR): 50 Marks
Prerequisite: Digital Electronics and Logic Design Laboratory.		

Course Objectives:
The objective of this course is to provide the students

1. Basic architectural features of processors.
2. Instruction set and logic to build assembly language programs.
3. Command line argument and stack simulation using tools.
4. Multithreading concepts using libraries.

Course Outcomes:
At the end of the course students will be able to
CO1:Exhibit skill of assembly language programming using instruction set.
CO2:Demonstrate stack simulation using function call.
CO3:Apply multithreading techniques to manage memory access.

List of Assignments

1.	Write inline assembly code for adding two short/long int numbers.
2.	Write inline assembly code to find signed multiplication and division.
3.	Write inline assembly code to accept the string and to display its length.
4.	Write inline assembly code to count the number of positive and negative numbers from the array.
5.	Simulation of stack operation in function call-with parameter using tool.
6.	Write inline assembly code to find the factorial of a given integer number on a command line.
7.	Write multi threaded application for Arithmetic calculators
8.	Write a multi threaded application to demonstrate shared memory access.

03051X1: Multidisciplinary Minor (MDM-1)		
Teaching Scheme: L: 02 Hrs./week	**Credits**: 02	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks
Refer : Multidisciplinary Minor Courses		

03051X1: Multidisciplinary Minor Tutorial/Lab (MDM-1)		
Teaching Scheme: T : 01 Hr./week	**Credit:** 01	**Examination Scheme:** CIE(TW): 25 Marks
Refer : Multidisciplinary Minor Courses		

1309101: Engineering Economics and Financial Management		
Teaching Scheme: L: 01 Hr./week	**Credit:** 01	**Examination Scheme:** CIE(TW): 25 Marks

Prerequisite: Basic Mathematics, Introductory Engineering Concepts.

Course Objectives:
The objective of this course is to provide the students

1. Concept of time and value of money.
2. Knowledge of financial evaluation techniques.

Course Outcomes:
At the end of the course students will be able to

CO1: Compute the worth of money at various points of time.
CO2: Apply various depreciation methods in determining the value of an asset.
CO3: Evaluate the replacement of an existing asset based on standard replacement analysis techniques.
CO4: Evaluate the best alternative in Engineering Economics problems considering risk and safety.

Course Contents

Module I	Financial Decision Making: Time Value and Cash Flow Analysis	4 Hrs.

Time value of money, Interest factors for discrete compounding, Nominal & effective interest rates, Present and future worth of Single, Uniform, and Gradient cash flow. Related problems and case studies.

Module II	Economic Analysis and Decision-Making Techniques	4 Hrs.

Bases for comparison of alternatives, Present worth amount, Capitalized equivalent amount, Annual equivalent amount, Future worth amount, Capital recovery with return, Rate of return method, Incremental approach for economic analysis of alternatives, Replacement analysis.

Module III	Rate-of-Return Analysis and Depreciation	4 Hrs.

Break even analysis for single product and multi product firms, Break even analysis for evaluation of investment alternatives. Physical & functional depreciation, Straight line depreciation, declining and double declining balance method of depreciation, Sum-of-the-Years Digits, Sinking Fund and Service Output Methods, Case Study.

Module IV	Project Risk and Uncertainty	3 Hrs.
Balance sheet and profit & loss statement. Meaning & Contents. Ratio analysis, financial ratios such as liquidity ratios, Leverage ratios, Turnover ratios, and profitability ratios, Drawbacks. Safety and Risk, Assessment of Risk and safety, Case study, Risk Benefit Analysis and Reducing Risk.		
Textbooks:		
T1. Chan S. Park, "Contemporary Engineering Economics", 4th Edition, Pearson Prentice Hall 2007, ISBN-978-0131492480. **T2.** Thuesen G. J, “Engineering Economics”, Prentice Hall of India, New Delhi,2005, ISBN- 978-0131233232. **T3.** Blank Leland T. and Tarquin Anthony J., “Engineering Economy”, McGraw Hill, Delhi, 2002, ISBN-978-0074620427.		
Reference Books:		
R1. Prasanna Chandra, “Fundamentals of Financial Management”, Tata McGraw Hill, Delhi-2006, ISBN-978-0070609137. **R2.** Mike W. Martin and Roland Schinzinger, “Ethics in Engineering”, Tata McGraw Hill, New Delhi- 2003, ISBN- 978-0074602751. **R3.** Govindarajan M, Natarajan S, Senthil Kumar V. S, “Engineering Ethics”, Prentice Hall of India, New Delhi- 2004,ISBN: 978-0131452412.		
Relevant Topics for Self-study:		
Micro, Small & Medium Enterprises (MSME) and its role in the Indian Economy.		

0308202: Professional Development and Career Readiness		
Teaching Scheme: P: 02 Hrs./week	**Credit**: 01	**Examination Scheme:** CIE(TW) : 25 Marks
Refer : Common Courses		

03063XX : OE-I Foreign Language Studies - German (FLSG)		
Teaching Scheme: T: 02 Hrs./Week	**Credits:** 02	**Examination Scheme:** CIE(TW): 50 Marks
Refer Annexure-II for guidelines		
Refer : Common Courses		

03063XX: : OE-I Foreign Language Studies - Japanese (FLSJ)		
Teaching Scheme: T: 02 Hrs./Week	**Credits:** 02	**Examination Scheme:** CIE(TW): 50 Marks
Refer Annexure-II for guidelines		
Refer : Common Courses		

0311101: Universal Human Values		
Teaching Scheme: L: 02 Hr./Week	**Credits:** 02	**Examination Scheme:** CIE(TW): 25 Marks
Refer : Common Courses		

03132XX: Community Engagement Project		
Teaching Scheme: P: 2 Hrs./ Week	**Credit:** 01	**Examination Scheme:** CIE(TW): 25 Marks
Refer : Common Courses		

Second Year B-Tech
(S. Y B. Tech)
Semester-4

1403106: Software Engineering		
Teaching Scheme: L: 02 Hrs./Week	**Credits**: 02	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Fundamentals of Programming Languages

Course Objectives:
The objective of this course is to provide the students

1. Concepts of software engineering and its diagrams.
2. Project planning and estimation.
3. Application of software modeling for problem solving.
4. Knowledge of risk analysis and software testing techniques.

Course Outcomes:
At the end of the Course Students will be able to:

CO1: Analyze software requirements.
CO2: Estimate and plan software projects effectively
CO3: Design modular and maintainable software systems using appropriate design processes, patterns, and core design concepts.
CO4: Identify the risk and Evaluate the software using software testing.

Course Contents

Module I	Introduction	7 Hrs.

Introduction to software engineering, The nature of software, Define software, software engineering practice, Software Process: A generic process model, A framework activity, Requirement Vs Analysis Vs Architecture Vs Design Vs Development 4+1 view Architecture, Introduction to UML -Basic building blocks.
Requirement Engineering: Building the requirements model. Agile Model.

Case study: Write Software Requirements Specification (SRS) in IEEE format for given problem statements.

Suggested List of Lab Assignment:Idea Inception for a given problem statement.

Module II	Planning in software engineering	6 Hrs.

Estimation: Project planning process, defining scope, checking feasibility, Resource management, Decomposition techniques, software sizing, LOC based & FP based Estimation. The COCOMO-II Model.Project Scheduling: Defining the task for project scheduling.

Case Study: Preparing requirement traceability matrix.

Module III	Modeling in Software Engineering	6 Hrs.

Design: Process, Attributes. Design Patterns: Creational Design Pattern, Structural Design Pattern, Behavioral Design Pattern.
Concepts: Abstraction, Architecture, Modularity, information hiding, functional independence, refactoring.

Case study- Design for a given problem statement.

Module IV	Risk Management and Deployment	7 Hrs.
Risk Management: Software risks, risk identification, risk projection, risk refinement, risk mitigation, monitoring and management (The RMMM plan). Software Configuration Management Testing: verification & validation, software testing strategies. **Case Study:** Selenium testing for given web application.		

Text Books:

T1. Roger Pressman, "Software Engineering: A Practitioner's Approach", McGraw Hill, ISBN-007–337597–7.
T2. Ian Sommerville, "Software Engineering", Addison and Wesley, ISBN- 0-13-703515-2.

Reference Books:

R1. S K Chang, ―Handbook of Software Engineering and Knowledge Engineering‖, World Scientific, Vol I, II, ISBN- 978-981-02-4973-1.
R2. Pankaj Jalote, ―An Integrated Approach to Software Engineering‖, Springer, ISBN- 978-8173192715.
R3. Carlo Ghezzi, ―Fundamentals of Software Engineering", Prentice Hall India, ISBN-10: 0133056996.
R4. Tom Halt, ―Handbook of Software Engineering‖, Clanrye International ISBN- 10: 1632402939.

Relevant Topics for Self-study:

Case Studies for various Applications like: Design a software system for an Automated Teller Machine (ATM) that allows bank customers to perform transactions such as cash withdrawal, balance inquiry, deposit, and PIN change. The system must authenticate users through a PIN, process transactions securely, update account balances in real-time, and provide appropriate messages or error notifications. The ATM should interface with the bank's central system for account verification and transaction processing.

1403107: Database Management System		
Teaching Scheme: L: 03 Hrs./Week	**Credits**: 03	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Discrete Mathematics, Data Structures

Course Objectives:
The objective of this course is to provide the students

1. Fundamental concepts of database management like database design and database languages.
2. SQL and NoSQL concepts through database management tools.
3. Basics of transaction management and concurrency control.

Course Outcomes:
At the end of the Course Students will be able to

CO1: Design an E-R Model for the given requirements and convert into normalized database tables.
CO2: Write SQL queries and PL/SQL code for given requirements.
CO3: Illustrate NoSQL database concepts and write queries using MongoDB.
CO4: Describe database transaction management concepts.

Course Contents

Module I	Introduction to Databases and Database Design	10 Hrs.

Introduction: Need for Database Management Systems, Evolution, Database System Concepts,and Architecture.
Data Modeling: Entity Relationship (ER) Model,Relational Model, Extended ER Model, converting ER and EER diagram into tables.
Database Design: Codd's Rules, Need of Normalization, Functional Dependencies, ~~Functional Dependency Closure, Decomposition Properties,~~ Normal Forms: 1NF, 2NF, 3NF, and ~~BCNF~~.

Module II	SQL and PL/SQL	7 Hrs.

SQL Characteristics and Advantages, Data Types and Literals, DDL, DML, Select Queries and clauses, SQL Operators, Functions, Aggregate Functions, Nested queries, Join Queries, **Views:** Creating, Dropping, Updating, Index and Sequence in SQL, DCL, TCL, **PL/SQL**: Procedure, Function, Cursors, Trigger.

Module III	NoSQL Databases	7 Hrs.

Introduction to NoSQL Databases, Types of NoSQL Databases, BASE properties, CAP Theorem, Comparative study of RDBMS and NoSQL, MongoDB (syntax and usage): CRUD Operations, Indexing, Aggregation, MapReduce.

Module IV	Database Transactions	7 Hrs

Introduction to query processing,basic concepts of a Transaction, ACID Properties, State diagram, Concept of Schedule.
Serializability – Conflict and View, Concurrency Control Protocols-Lock based and timestamp-based

protocols, Recovery techniques.

Text Books:
T1. Avi Silberschatz, Henry F. Korth, S. Sudarshan, Database System Concepts, 2019, Seventh Edition, McGraw-Hill, ISBN 9780078022159.
Reference Books:
R1. Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, "MongoDB: The Definitive Guide", 2020, Third Edition, O'Reilly Publications, ISBN- 978-1-449-34468-9. **R2.** S. K. Singh, "Database Systems: Concepts, Design and Application", 2013, Second Edition, Pearson Education, ISBN- 978-81-317-6092-5.
Relevant Topics for Self-study:
Inference Rules, Advanced nested queries in SQL.
Relevant MOOC Course:
NPTEL Lecture Videos by Dr. Arnab Bhattacharya from IIT Kanpur. NOC:Fundamentals of Database Systems - http://www.nptelvideos.com/course.php?id=426

1403108: Operating Systems		
Teaching Scheme: L: 02 Hrs./week	**Credits**: 02	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Computer Fundamentals

Course Objectives:
The objective of this course is to provide the students

1. OS internals and functioning.
2. Organization of data and I/O devices in OS.
3. Mechanism of interacting user space applications with kernel space,Linux system calls.
4. Shell scripting constructs.

Course Outcomes:
At the end of the course students will be able to

CO1: Analyse various strategies of Process and Memory management in OS.
CO2: Compare different file systems and device management methods.
CO3: Develop applications using system calls and containerization.
CO4: Write shell scripts for managing users and resources.

Course Contents

Module I	Introduction, Process and Memory Management	10 Hrs.

Introduction: Introduction to Operating Systems, definition, functions, and types of operating systems, OS Structure: Monolithic, Microkernels, Layered, Modular, and Hybrid systems, OS Services, and Design, Modern Operating Systems and Concepts like containerization, virtualization and Dockers.

Process Management: Process Concept: States, Process Control Block (PCB), **Multithreading:** Threads, User vs Kernel Threads, CPU Scheduling: SchedulingCriteria, Types of Schedulers, Scheduling Algorithms: FCFS, SJF, Priority, Round-Robin, Inter-process Communication (IPC): Process Synchronization, Critical Section Problem, Semaphores, and Monitors, Philosophers), Deadlock:Detection, Prevention, and Recovery. Multi threaded echo server.

Memory Management: Memory Management Requirements, Memory Partitioning, and Virtual memory , Placement Algorithms: First Fit, Best Fit,Worst Fit, Paging, Page replacement algorithms: FIFO, LRU, OPTIMAL,Segmentation.

Module II	File Systems and Device Management	8 Hrs.

File Systems: Overview, File Organization and Access, File Directories, File Sharing, Record Blocking, Secondary Storage Management, File System Security, UNIX File System, Linux Virtual File System, Windows File System, Case studies of advanced file systems (ext4, ZFS, Btrfs).

Device Management: I/O Devices, Organization of the I/O Function, Operating System Design Issues, I/O Buffering, I/O hub, Disk Scheduling Policies: FIFO, PRIORITY, SSTF, SCAN, C-SCAN, Linux I/O, Windows I/O.

Module III	Systems Calls Design and Implementation	8 Hrs.

Introduction to System Calls: Definition of system calls and their role in the OS, Differences between user mode and kernel mode, Overview of the system call interface.

Mechanisms of System Calls: Working of a system call (traps, interrupts),Transition from user mode to kernel mode, Context switching and its overhead.

Types of System Calls: Process Control: Creating and terminating processes (fork, exec, exit), Process synchronization (wait, signal), Process scheduling and priority management.

File: File creation, deletion, and manipulation (open, read, write, close), File attributes and permissions, Directory operations (mkdir, rmdir,chdir).

Error Handling: Common errors encountered with system calls, Error codes and how to interpret them, Best practices for robust error handling.

Module IV	Shell Scripting	8 Hrs.

Introduction to Shell Scripting: Overview of shell scripting and its purpose,Types of shells (e.g., Bash, Zsh, Ksh, Csh), Differences between command line and shell scripts.

Basic Shell Commands: Common Unix/Linux commands (ls, cp, mv, rm, cat,echo), Command-line arguments and options, Understanding standard input,output, and error streams.

Writing First Shell Script: Script structure and syntax,Creating and executing a simple shell script, Using the shebang (#/!) for specifying the interpreter. User management.

Variables and Data Types: Declaring and using variables, Variable scope and export, Special variables.

Control Structures: Conditional statements (if, else, elif,case), Looping constructs (for, while, until), Using break and continue in loops.

Functions: Defining and calling functions, Passing arguments to functions, Returning values from functions.

Input and Output: Reading user input with read, Redirecting input and output, using pipes to connect commands.

Text Books:

T1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006), "Operating System Principles", 8th edition, Wiley India Private Limited, New Delhi, ISBN: 978-8126509621.

T2. William Stallings (2006), Operating Systems, Internals and Design Principles, 9th edition, Pearson Education, India, ISBN: 978-9352866717.

T3. Maurice J. Bach, The Design of the UNIX Operating System, Prentice-Hall, 8th edition, Prentice-Hall software series, ISSN: 978-0132017991.

Reference Books:

R1. Andrew S. Tanenbaum (2007), "Modern Operating Systems", 3rd edition, Prentice Hall of India, ISBN-978-8120339040.

R2. Deitel & Deitel (2008), "Operating systems" , 3rd edition, Pearson Education, India, ISBN-978-0131828278.
Relevant Topics for Self-study:
I/O management, Interrupt handling, Device drivers, Daemons, Dynamic Link Libraries concepts in Windows/Ubuntu/ Android, etc, containerization of an application.
Relevant MOOC Course:
1. NPTEL course on, "Operating Systems Fundamentals", Prof. Santanu Chattopadhyay, IIT Kharagpur **https://www.google.com/url?q=https://onlinecourses.nptel.ac.in/noc24_cs108/preview&sa=D&source=editors&ust=1727676932469729&usg=AOvVaw3r43hMlIHD5KfopuUmaHJl**
Other Resources/Links:
1. https://www.opensourceforu.com/ 2. https://www.minix3.org/

1403209: Operating Systems Lab		
Teaching Scheme: L: 02 Hrs./Week	**Credit**: 01	**Examination Scheme:** CIE(TW): 25 Marks ESE(OR): 25 Marks

Prerequisite: Fundamentals of programming

Course Objectives:
The objective of this course is to provide the students

1. Process and memory management techniques.
2. Disk scheduling and internals of storage management.
3. Usage of various system calls and shell scripting.

Course Outcomes:
At the end of the course students will be able to

CO1: Implement and **analyse** different scheduling algorithms FCFS,SJF,RR and priority.
CO2: Apply different memory placement and replacement strategies for a given input requirements.
CO3: Demonstrate various internals of the Linux file system.
CO4: Write the shell scripts to manage the users and access rights.

List of Assignments

1.	**Assignment 1:** Containerization of a Web Application using Docker **Objective:** Containerize a simple web application using Docker, understand the container lifecycle, and explore best practices for optimizing containerized deployments. **Problem Statement:** Develop a lightweight and portable web application that needs to run consistently across multiple environments (development, testing, and production). Following are the functionalities you need to perform: i. Containerize a simple Python Flask web application using Docker. ii. Ensure that the application runs consistently across different environments. iii. Optimize the container by reducing image size and improving performance. iv. Deploy and manage the container using essential Docker commands.
2.	**Assignment 2:** Microkernel Simulation **Problem Statement: :** Write a simple microkernel simulation using C/Python to: i. Handle messages passing between system services. ii. Simulate basic IPC (Inter-Process Communication). iii. Run the program and demonstrate IPC functionality.
3.	**Assignment 3:** GRUB Configuration & Hard Disk Partitioning. **Problem Statement:** Your organization is setting up a multi-boot environment with Linux and Windows, requiring proper GRUB configuration and disk partitioning for smooth operation. You have been assigned to: i. Analyze the GRUB boot process and its role in loading the OS. ii. Customize the GRUB menu, setting the default OS, timeout, and adding custom boot parameters. iii. Simulate a boot failure scenario and apply recovery techniques to restore GRUB. iv. Partition a hard disk, implementing best practices for Linux installation, including swap space and EFI partition. v. Perform a dual-boot setup with Linux and Windows, ensuring proper bootloader functionality. vi. Document troubleshooting steps for common GRUB issues, such as missing entries or boot loops.

1403210: Database Management System Lab		
Teaching Scheme: P: 04 Hrs./week	**Credits**: 02	**Examination Scheme:** CIE(TW): 25 Marks ESE(PR): 25 Marks
Prerequisite: Database Management System		

Course Objectives:

The objective of this course is to provide the students

1. Knowledge to handle SQL statements and PL/SQL code in databases.
2. Skills to handle NoSQL databases.
3. Application development with database connectivity.

Course Outcomes:

At the end of the course students will be able to

CO1: Implement the given relational schema, database queries, and PL/SQL programs for 2-tier architecture using MySQL.

CO2: Implement NoSQL queries and aggregate functions for given requirements using MongoDB.

CO3: Develop database applications using database connectivity.

List of Assignments

1.	Use an open source database tool to implement a structured database system for a Retail Store to manage products, customers and orders. Write SQL queries to perform DDL operations for table creation in the Retail Store database with constraints (Not Null, Primary, Foreign etc) and Modification using Alter command variations, and also perform basic DML operations (Insert, Select, Update, Delete), and executing various Select statement queries with different clauses for given schemas of Retail Store database.
2.	An University maintains a structured database to manage student enrollments, courses, instructors and departments. Write SQL queries using various JOIN types (Natural, Inner, Equi, Non-Equi, Outer, Left Outer, Right Outer), as well as Subqueries and Views to extract insights from the university database system.
3.	A Bank wants to automate its customer credit scoring system based on their financial standing. Let the credit score be computed on a scale of 0 to 5 by a PLSQL function based on the account balance and loan amount of any customer. (hint: if loan_amount is 50% of account balance then, credit_score: 3). Write a stored procedure that uses a cursor and calls the function with each customer name for computing credit scores of all bank customers. i. Customer(Cust_name, AccNo, Balance, city) ii. Loan(Loan_no, branch_name, Amount) iii. Borrower(Cust_name, Loan_no, CreditScore)
4.	A Company wants to track employee salary changes, maintain company-wide statistics, and log employment history efficiently using row-level triggers. Implement BEFORE and AFTER triggers on EMPLOYEE, COMPANY_INFO, and EMP_LOG tables using INSERT, UPDATE and DELETE operations.

	i. EMPLOYEE(Emp_Id, First_Name, Last_Name, Email, Phone_No, Hire_Date, Job_Profile, Salary, HRA) ii. COMPANY_INFO(Emp_Count, Total_Salary_Expenses) iii. EMP_LOG(Emp_Id, Old_Salary, New_Salary, Edit_Time, Job_Status)
5.	An institute maintains details of all teachers, including name, qualifications, department details, experience, salary structure, date of joining, appointement_nature and area of expertise. Design and implement MongoDB queries to perform CRUD operations on the teachers collection for various administrative tasks. Create the above collection, insert suitable documents and design updation and retrieval queries requiring comparison and logical operators, save() method, etc.
6.	Customers of an online marketplace frequently search for products and their orders, and the search becomes slow as the products and orders grow. Implement all types of indexes on the products and order MongoDB collections.
7.	The University wants to analyze course enrollments, faculty performance and student achievements using MongoDB aggregation pipelines. The university maintains a "courses" collection, which includes: i. Course details (title, department, credit hours, instructor) ii. Student enrollments (student names, scores, pass/fail status) iii. Faculty details (experience, designation, department) Design and implement aggregation queries to generate various reports using multiple aggregation stages with the given collection in MongoDB.
8.	Implement various database operations through suitable programming language.

04051X2: Multidisciplinary Minor (MDM-2)		
Teaching Scheme: L: 02 Hrs./week	**Credits**: 02	**Examination Scheme:** CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks
Refer : Multiltidisciplinary Minor Course		

04052X2:Multidisciplinary Minor (MDM-2)		
Teaching Scheme: P : 02 Hrs./week	**Credit:** 01	**Examination Scheme:** CIE (TW) : 25 Marks
Refer : Multidisciplinary Minor Course		

04063XX: NPTEL / SWAYAM / MOOCs		
Teaching Scheme: Online 2 Hrs/ Week	**Credits:** 02	**Evaluation Scheme** ESE: 50 Marks
Refer : Annexure-II for guidelines		

1407201: Project Based Learning		
Teaching Scheme: P: 04 Hrs./week	**Credits**: 02	**Examination Scheme:** ESE(OR) : 50 Marks

Prerequisite: 1. Basic Subject Knowledge: Foundational understanding of the technology stack
2. Design thinking lab

Course Objectives:
The objective of this course is to provide the students

1. Independent learning by multi-disciplinary problem-solving with social context.
2. Opportunity to get involved either individually or as a group so as to develop team skills and learn professionalism.

Course Outcomes:
At the end of the course students will be able to

CO1: Understand project lifecycle with relevance to real-world issues.
CO2: Apply development /design skills in the project in a practical context.
CO3: Analyze different complex problem breakdown processes.
CO4: Assess the effectiveness of the methods being used throughout the project.
CO5: Design and develop a comprehensive project that addresses a specific societal need.

Suggested Problem Statements(Not limited to this choose any)		
Sr No	Domain	Sample Problem Statement
1	Waste Management Optimization	Inefficient waste collection leads to overflowing bins and increased pollution.
2	Renewable Energy Monitoring	Limited visibility on renewable energy production affects efficiency.
3	Urban Air Quality Monitoring	Poor air quality in urban areas poses health risks.
4	Water Quality Testing	Contaminated water sources threaten public health.
5	Smart Traffic Lights	Traffic congestion causes delays and pollution.
6	Elderly Care Monitoring System	Elderly individuals living alone face health risks and emergencies.
7	Food Safety Tracking	Problem: Foodborne illnesses due to unsafe food handling practices.
8	Energy Consumption Management	Problem: High energy consumption leads to increased costs and carbon footprint.
9	Disaster Response App	Problem: Ineffective communication during natural disasters.

10	Personalized Learning Tools	Problem: One-size-fits-all education fails to meet diverse student needs.

Indicative/ Sample List of Tools (Students are allowed to use any latest tools / Not Limited to this):

1. Git/GitHub - Version control and collaboration.
2. Visual Studio Code - Integrated development environment (IDE).
3. AWS/Google Cloud/Azure - Cloud computing platforms.
4. TensorFlow/PyTorch/Scikit-learn - Machine learning libraries.
5. React/Angular/Django - Web development frameworks.

Text Books:

T1. Michael E. Auer "Project-Based Learning in the Computer Science Classroom", ISBN- 978-3-319-70663-0.
T2. Richard DuFour, Rebecca DuFour, and Robert Eaker "Learning by Doing: A Handbook for Professional Learning Communities at Work" , Solution Tree, ISBN- 978-1-4166-0732-8.
T3. Gary R. Morrison, Steven M. Ross, and Jerrold E. Kemp "Designing Effective Instruction" 6th Edition , John Wiley & Sons.ISBN- 978-1-118-20763-4.
T4. Nell Dale and John Lewis "Computer Science Illuminated" Jones & Bartlett Learning, 7th Edition ISBN- 978-1-118-45306-7.

Paper References:

1. Junior, E., et al. "Systematic literature review of Gamification and Game-based Learning in the context of Problem and Project Based Learning approaches." International Symposium on Project Approaches in Engineering Education. Vol. 9. 2019.
2. Malik, Khalid Mahmood, and Meina Zhu. "Do project-based learning, hands-on activities, and flipped teaching enhance student's learning of introductory theoretical computing classes?." Education and information technologies 28.3 (2023): 3581-3604.
3. Faizi, Jamilurahman, and Mohammad Sarosh Umar. "A conceptual framework for software engineering education: project based learning approach integrated with industrial collaboration." International Journal of Education and Management Engineering 11.5 (2021): 46.

Relevant MOOCs:

1. NPTEL course on, "Compiler design", By Prof. Santanu Chattopadhyay, IIT Kharagpur https://onlinecourses.nptel.ac.in/noc21_cs07/preview

1409102: Entrepreneurship		
Teaching Scheme: L:01 Hr./week	**Credit:** 01	**Examination Scheme:** CIE :25 Marks

Prerequisite: Problem-solving abilities, Leadership skills

Course Objectives:
The objective of this course is to provide the students

1. Skills to identify target customers, segment markets.
2. Utilization of tools for designing and developing effective business models.
3. Understanding of team dynamics and core business principles.

Course Outcomes:
At the end of the course students will be able to

CO1. Design innovative and sustainable business models.
CO2. Develop business strategies to foster business development. / Establish Business by considering all the requirements.

Course Contents

Module I	Entrepreneurship in Practice	8 Hrs.

Team building, Shared leadership, role of good team, Collaboration tools and techniques, Marketing and sales, Positioning, Channels and strategies, Sales planning,Support, Project management, Planning and tracking, Business Regulation.Customer Segmentation.

Module II	Economics and Management for Entrepreneurs	7 Hrs.

Customer identification, Market, Creative solution, Unique Value proposition Business Model Canvas, Types of business models, Risk identification, Business Plan documentation, Business Establishment Requirements.

Textbooks:

T1. Robert D. Hisrich,, “Advanced Introduction to Entrepreneurship”, Elgar Advanced Introductions series, Extent: 296 pp, ISBN- 978- 1 78254 6153.
T2. Henry Rwigema and Robert Venter, “Advanced Entrepreneurship”, Oxford University Press, 2004 ISBN- 978- 0195780574.

Reference Books :

R1. Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd, Sabyasachi Sinha, "Entrepreneurship", 11th Edition, Published: July 20, 2020, ISSN- 978-9390113309.
R2. Barbara J. Orser and Catherine J. Elliott, “Feminine Capital: Unlocking the Power of Women Entrepreneurs”, Stanford University Press, Edition: 1st, DOI:10.1515/9780804794312, ISBN- 978-0804794312.

Relevant Topics for Self-study:

Case studies on Leadership and Market strategies.

1409202: Entrepreneurship Lab		
Teaching Scheme: P: 02 Hrs./week	**Credit:** 01	**Examination Scheme:** CIE (TW) : 25 Marks
Prerequisite: Problem-solving abilities, Leadership skills		

Course Objectives:
The objective of this course is to provide the students

1. Explore the intersection of innovative strategies, economic principles, and management practices to drive entrepreneurial success.
2. Ability to analyze market dynamics and craft innovative strategies for sustainable business growth in a competitive landscape.
3. Tools and techniques for effective decision-making, resource allocation, and team management in entrepreneurial ventures.
4. Incorporating sustainability and inclusivity into their entrepreneurial and managerial strategies, aligning with global economic trends.

Course Outcomes:
At the end of the course students will be able to

CO1. Demonstrate the ability to design and implement innovative strategies that integrate economic and managerial principles to solve complex business challenges.

CO2. Analyze market trends, economic policies, and business environments to identify opportunities and mitigate risks.

List of Assignments

1.	Create a report explaining key factors influencing the success or failure of market entry strategies for new entrepreneurs.
2.	Create and Present Your Own Business Model Canvas to outline the critical components: Value Proposition, Customer Segments, Channels, Customer Relationships, Revenue Streams, Key Resources, Key Activities, Key Partnerships, and Cost Structure.
3.	Create and Present a report of your own business with an analysis of the business's impact on local economics, considering both positive and negative externalities, and proposing strategies for improvement.

0408203: Collaborative Skills, Digital Ethics, and Cyber Security		
Teaching Scheme: P: 02 Hrs./week	**Credit**: 01	**Examination Scheme:** CIE(TW) : 25 Marks
Refer : Common Courses		

0411102: Indian Constitution and Social Responsibility (ICSR)		
Teaching Scheme: L: 01 Hrs./Week	**Credits**: 01	**Examination Scheme:** CIE: 25 Marks
Refer : Common Courses		

04132XX: Community Engagement project		
Teaching Scheme: P: 2 Hrs./ Week	**Credit:** 01	**Examination Scheme:** CIE(TW): 25 Marks
Refer : Common Courses		

Annexure-1

Structure of Multi-Disciplinary Minor Courses

The structure for the multidisciplinary Minor courses is as follows.

			Teaching Scheme (Hours/Week)				Credits				Examination Scheme and Marks						
											Theory			Practical			
Sem	Course code	Name of the Course (Short forms)	L	P	T	Total	L	P	T	Total	CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE (PR)	ESE (OR)	Total
3	03051X1	MDM-1	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
3	03052X1	MDM-1 Tut	-	-	1	1	-	-	1	1	-	-	-	25	-	-	25
4	04051X2	MDM-2	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
4	04052X2	MDM-2 Lab	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
5	05051X3	MDM-3	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
5	05052X3	MDM-3 Lab	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
6	06051X4	MDM-4	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
6	06052X4	MDM-4 lab	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
8	08053X5	MDM-5	-	-	2	2	-	-	2	2	-	-	-	50	-	-	50
		Total	8	6	3	17	8	3	3	14	80	80	240	150	0	0	550

Note: In course code X is basket number. #: is laboratory or tutorial as per course requirements.

1. Students are instructed to select one basket labeled as MD-1 to MD-9 based on applicable criteria in the last column of the table below.
2. Follow the registration process initiated by the institute/ department.
3. Once the course is registered, change of the basket/course in any of the further semesters is not allowed.
4. The total credits will be the same for all the courses except MDM-5.
5. Departments may conduct MDM-5 in virtual mode for the smooth conduction of internship for the full duration.

List of Multi-Disciplinary Minor Domains

Sr. No.	Multi-Disciplinary Minor Domains	SY		TY		B-Tech	Offered to students of BTech Program
		MDM-1	MDM-2	MDM-3	MDM-4	MDM-5	
		Sem-III	Sem-IV	Sem-V	Sem-VI	Sem-VII/VIII	
MD1	**Smart and Sustainable Systems (SSS)**	Fundamentals of Smart and Sustainable Systems (FS&SS) Tut.	IoT for Smart and Sustainable Systems (ISSS) & Lab	Data Analytics for Smart and Sustainable Systems (DASSS) & Lab	Security for Smart and Sustainable Systems (SSS) & Smart and Sustainable Systems Development (SSD) Lab	Smart and Sustainable System (SSD) (MOOC)	**ALL**
MD2	**Financial Technology and Management (FTM)**	Finance and Management (FM)	Fundamentals of Financial Engineering (FFE) & Tut	Banking, Financial Services and Insurance (BFSI) & Tut	Fundamentals of Stock Market (FSM) & Tut	Fintech: Foundations & Applications (FFA) & Tut	**ALL**
MD3	**3D- Printing (3DP)**	3D- Printing (3DP)	3D modeling and Design (3MD) & Lab	Fundamentals of Additive Manufacturing (FAM)& Lab	3D Printing Materials and Processes (3DPMP)	Industry 4.0 and Digital Manufacturing (IDM)	**ALL**
MD4	**Electric Vehicles (EV)**	Electric Vehicles (EV)	EV foundation – Principles and Concepts (EVPC) & Lab	Advanced Motor Technologies and Power Electronics for EV(AMT) & Lab	EV Powertrain Dynamics and Control System(PDC) Tut/Lab	Intelligent EV Systems: AI IoT and Automation(IEV)	**ALL**
MD5	**Applied Mathematics for Engineering (AME)**	Applied Mathematics for Engineering (AME)	Linear Algebra with Python & Lab	Statistical Techniques and Numerical Methods with R & Lab	Fuzzy Logic and Graph Theory with Matlab/Python & Lab	Optimization Techniques & Lab	**ALL**
MD6	**Software Development (SD)**	Software Development (SD)	Data Structures and Algorithms (DSA) & Lab	Object Oriented Programming (OOP) &Lab	Database and Management Systems (DBMS) & Lab	Web Development (WD) & Lab	**E&TCE**
MD7	**Autonomous and Intelligent Systems (AIS)**	Autonomous and Intelligent Systems (AIS)	Digital Systems and Organization (DSO) & Lab	Smart System Engineering (SSE) & Lab	Embedded IoT Systems (EIS) & Lab	Autonomous Systems (AS) & Lab	**All except E&TCE**
MD8	**Embedded Systems- (ES)**	Embedded Systems (ES)	Fundamental of Microcontroller (FM) & Lab	Embedded Processors –I (EP -I) & Lab	Microcontrollers and IoT (MI) & Lab	Embedded Systems and RTOS (ES-RTOS) & Lab	**All Except E&TCE**
MD9	**AI & Machine Learning (AI-ML)**	AI & Machine Learning (AI-ML)	Statistical Data Analysis & Lab	Machine Learning (ML) & Lab	Natural Language Processing (NLP) & Lab	Artificial Intelligence (AI) & Lab	**E&CE**

Link: Detailed Syllabus

Annexure -2

Guidelines for Open elective Courses

- Open Elective – I will be offered in the third semester as foreign language as prescribed in the structure.
- Open Electives – II, III, IV will be offered through SWAYAM/NPTEL MOOCs of Equivalent Credits.
- Departments will prepare the baskets/list of courses to be offered at the start of every semester, Students may choose any one course from the basket with adhering to any one stream.
- The department shall appoint a faculty mentor to ensure monitoring and course registration smoothly. The load for faculty should be reflected in the time table.
- Equivalent Teaching Scheme, credits and examination schemes are detailed in table below.
- End semester Evaluation: No separate examination is conducted by the institute.

			Teaching Scheme (Hours/Week)				Credits				Examination Scheme and Marks						
											Theory			Practical			
Sem	Course code	Name of the Course	L	P	T	Total	L	P	T	Total	CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE P	ESE OR	Total
3	OE-I	Foreign Language Studies (FLS)	-	-	2	2	-	-	2	2	-	-	-	50	-	-	50
4	OE-II	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50
5	OE-III	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50
6	OE-IV	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50

Guidelines for MOOCs

1. The department shall release a list of approved SWAYAM-NPTEL courses before the commencement of every semester.
2. Students shall register for the approved Courses as per the schedule announced by SWAYAM-NPTEL.
3. A student shall undergo the courses only from the list notified by the department through SWAYAM/NPTEL platform and complete all the assignments and examination requirements as specified by SWAYAM/NPTEL.
4. SWAYAM-NPTEL Courses are considered for transfer of credits only if the student concerned has

successfully completed and obtained the SWAYAM-NPTEL Certificate.

5. The credit equivalence for SWAYAM-NPTEL Courses: 12 weeks – 3 credits; 8 weeks – 2 credits; 4 weeks – 1 credit.
6. Equivalent marks will be considered for awarding the grades as specified in examination rules and regulations. The weightage for assignments is 40%, while the weightage for the proctored examination will be 60% for award calculating SGPA/CGPA. Students must score a minimum of 40% of the total marks by combining both assignments and proctored examinations
7. A student must submit the original SWAYAM-NPTEL Course Certificates to the Head of the Department concerned, with a written request for the transfer of the equivalent credits. On verification of the SWAYAM-NPTEL Course Certificates and approval by the head of the department, credits will be awarded.
8. The Institute shall not reimburse any fees/expenses a student may incur for the SWAYAM-NPTEL Courses.
9. If the SWAYAM/NPTEL course calendar does not align with the institute's calendar, the department shall facilitate and conduct examination of the relevant course of equivalent credits in physical/virtual mode and award the credits accordingly.

Guidelines for Evaluation

Continuous Internal Evaluation: (Weightage for Attendance: 5, Activity Based Learning Evaluation: 15)

The department shall declare the set of all applicable activities such as Problem Based Learning, Quizzes, Small Project, field work, group discussion, but not limited to etc. The course coordinator, in consultation with course teachers, shall select any two activities suitable for the course from the list declared by the department and get the selected activities approved from HoD. The Course teacher shall get the activities carried out by students, evaluate the student performance based on the prescribed rubrics. The department shall prepare the rubrics for all the activities and display the same before the commencement of academics.

In-Semester Examination: Written examination shall be conducted for one hour duration on First Module for 20 marks. End-Semester Examination: Written examination shall be conducted for two- and half-hour duration on Modules II, III, IV for 60 marks.

Annexure -3

Relevant topics for self study:

These topics are optional and suggested for students for self study to get additional exposure. However, these topics will not be considered for any evaluation and credits.